IoT Base Automated Street Lighting System

by
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CPE162P-4_E01
EMBEDDED SYSTEM WITH IOT

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Objective

Objective Overview:

- **To demonstrate how an IoT based Automated Street Lighting System works.**
- **To be able apply what you have learned in logic circuits and switching theory, microprocessors, CPE160P, and CPE161P.**
- **To be able to apply embedded system with IoT using Arduino kit**
- **To be able to use Thingspeak for the data collection and analysis.**

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EXPERIMENT OVERVIEW

- Components Used:
  - Arduino Uno
  - 10 Led
  - Photoresistor
  - 10k resistor
  - Wifi module

- Street light Feature:
  - Automatic On/Off depending on sunlight present
  - Real time data collection

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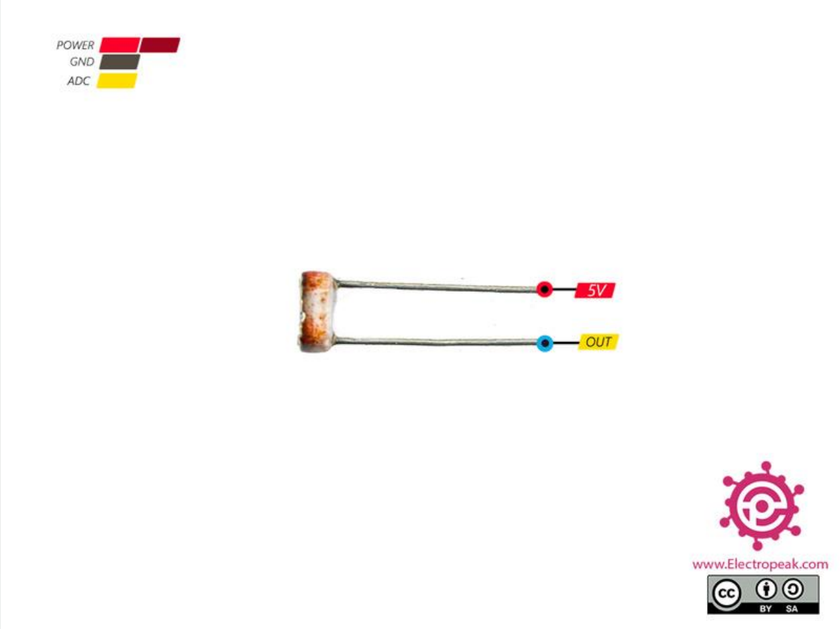
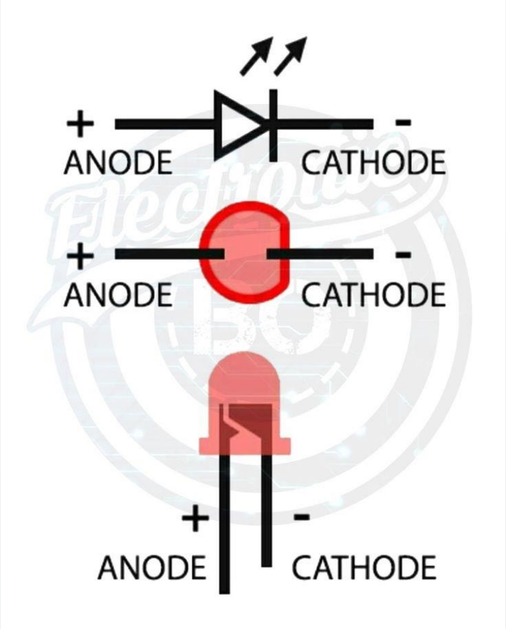
DISCUSSION - MATERIALS PIN OUT

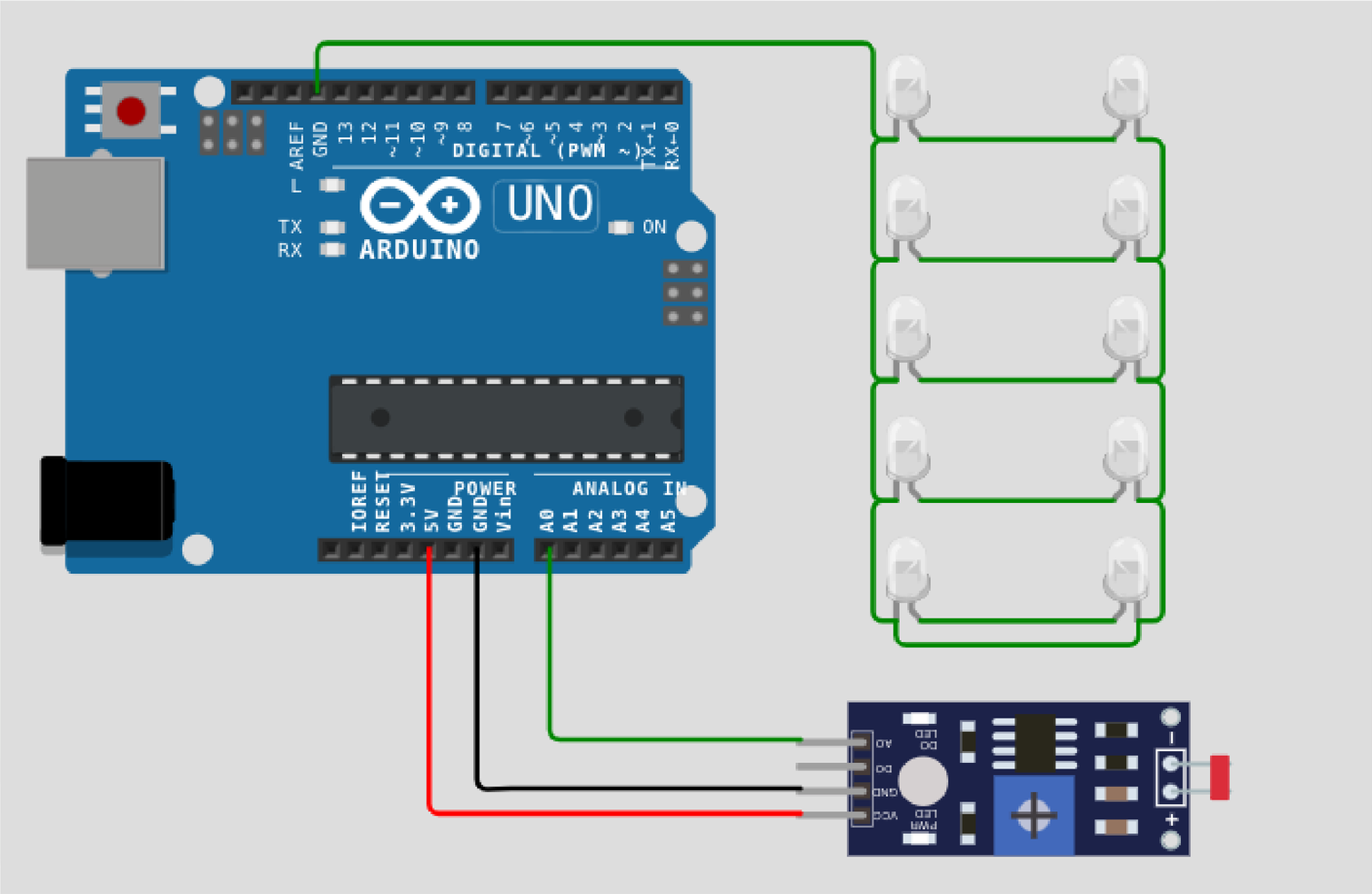


PHOTORESISTOR



led

DISCUSSION - CIRCUIT DIAGRAM



DISCUSSION - PHYSICAL IMPLEMENTATION & CODING

- **Circuit Construction:**
  - **Transferred circuit diagram created in simulation to a physical breadboard.**
  - **Assembled components: Led and photoresistor**

Code Construction:

  - **Develop code for the integration of experiment in the simulation then transfer**
  - **Test the functionality of each component**

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DISCUSSION - INITIALIZATION

```
#include <SoftwareSerial.h>

// Define pins
#define LDR_PIN A0
#define LED1 2
#define LED2 3
#define LED3 4
#define LED4 5
#define LED5 6

// ESP8266 TX to Arduino pin 10, RX to pin 11
SoftwareSerial esp(10, 11);

// ThingSpeak API key
String apiKey = "GVGISNVR0G07166P";

// Wi-Fi credentials
const char* ssid = "Ryna";
const char* password = "Ryna1234";
```

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DISCUSSION - Pin Setup

```
void setup() {
  Serial.begin(9600);
  esp.begin(115200);

  pinMode(LDR_PIN, INPUT);
  for (int i = 2; i <= 6; i++) pinMode(i, OUTPUT);

  connectWiFi();
}
```

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DISCUSSION - Loop

```
void loop() {
  int lightVal = analogRead(LDR_PIN);
  Serial.print("Light Level: ");
  Serial.println(lightVal);

  int ledStatus = 0; // 0 - OFF, 1 - ON

  // LED Control Logic
  if (lightVal < 400) { // If it's dark, turn on LEDs
    for (int i = 2; i <= 6; i++) {
      digitalWrite(i, HIGH);
    }
    ledStatus = 1; // LEDs are ON
  } else { // If it's bright, turn off LEDs
    for (int i = 2; i <= 6; i++) {
      digitalWrite(i, LOW);
    }
    ledStatus = 0; // LEDs are OFF
  }

  // Upload both LDR value and LED status to ThingSpeak
  uploadToThingSpeak(lightVal, ledStatus);
  delay(15000); // ThingSpeak requires 15s delay between uploads
}
```

DISCUSSION – ThingSpeak connectivity

```
void connectWiFi() {
  sendCommand("AT", 2000);
  sendCommand("AT+CWMODE=1", 2000);
  sendCommand("AT+CWJAP=\"" + String(ssid) + "\",\"" + String(password) + "\"", 6000);
}

void uploadToThingSpeak(int lightVal, int ledStatus) {
  String cmd = "AT+CIPSTART=\"TCP\",\"api.thingspeak.com\",80";
  sendCommand(cmd, 2000);

  String getStr = "GET /update?api_key=" + apiKey + "&field1=" + String(lightVal) + "&field2=" + String(ledStatus) + "\r\n";
  sendCommand("AT+CIPSEND=" + String(getStr.length() + 2), 2000);
  esp.print(getStr);
}

void sendCommand(String command, int timeout) {
  esp.println(command);
  long int time = millis();
  while ((time + timeout) > millis()) {
    while (esp.available()) {
      Serial.write(esp.read());
    }
  }
}
```

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DISCUSSION - INTEGRATION AND TESTING

- **Test Automative Led Functionality:**
  - Verify if right LDR value will turn on and off LED connectivity.
- **Test ThingSpeak Data Gathering Functionality:**
  - Verify if Led status corresponding with the LDR Value reflect in ThingSpeak

CHALLENGES

CHALLENGES

- Connection of Arduino wifi module
- Connection of wifi module to thingspeak

CONCLUSION

Objectives Achieved

- Successfully demonstrated an IoT-Based Automated Street Lighting System using Arduino and ESP8266.
- Applied logic circuits, switching theory, and embedded system concepts from CPE160P and CPE161P.
- Integrated ThingSpeak to collect, store, and analyze real-time LDR and LED state data.
- Built a system that responds to light conditions automatically—LEDs turn ON when light < 400, OFF otherwise.

Key Learnings

- Gained hands-on experience with ESP8266 WiFi module setup and Arduino communication.
- Understood the basics of cloud-based IoT platforms (ThingSpeak) for monitoring and data visualization.
- Learned how to send and update data periodically to the cloud for real-time analysis.
- Strengthened debugging skills and problem-solving during WiFi and cloud integration challenges.
- Developed a deeper understanding of sensor-based automation systems.

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